Computer Science - K-12 Level Assessment 12 Total Marks: 40 Answer all questions.

# Multiple Choice (10 marks)

1. Which of the following has the greatest potential for compromising a user's personal privacy?
   (a) A group of cookies stored by the user's Web browser
   (b) The Internet Protocol (IP) address of the user's computer
   (c) The user's e-mail address
   (d) The user's public key used for encryption
2. What is the value of this python expression: 4 + 3 % 2?
   (a) 1
   (b) 6
   (c) 4
   (d) 5
3. In Python 3, b = [11,13,15,17,19,21]; print(b[::2]) outputs what?

19,21

11,15

11,15,19

13,17,21

4. Which of the following is the hexadecimal representation of the decimal number 231_{10} ?
   (a) 17_{16}
   (b) $E_4$_{16}
   (c) $E_7$_{16}
   (d) $F_4$_{16}
5. Which is the smallest asymptotically?
   (a) O(1)
   (b) O(n)
   (c) O($n^2$)
   (d) O(log n)

# True / False (10 marks)

*Answer True or False*

6. True or False: In Python 3, floor division is performed using the operator '/'.
7. True or False: Authentication is compromised when an unauthorized user logs into a system using stolen credentials.
8. True or False: A distributed denial-of-service (DDoS) attack involves multiple computers launching the attack simultaneously.
9. True or False: In Python, the expression list[1:3] with list = [ 'abcd', 786 , 2.23, 'john', 70.2 ] results in [786, 2.23].
10. True or False: Python variable names are case-sensitive, meaning 'Variable' and 'variable' are treated as different identifiers.

# Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Write a function to find the number of possible sequences of length n such that each of the next element is greater than or equal to twice of the previous element but less than or equal to m.
12. Write a function to perform the adjacent element concatenation in the given tuples.

*End of Paper*